Protection Module for Water and Waste Water Products

AmaControl

Type Series Booklet

Legal information/Copyright

Type Series Booklet AmaControl

Contents

Monitoring and Diagnostic Systems

Protection Module for Water and Waste Water Products

AmaControl

Main applications

- Protecting and monitoring pumps, pump systems and submersible mixers in the most diverse of applications
- Diagnosing pumps, pump systems and submersible mixers to ensure trouble-free and reliable operation
- Analysing and linking data for condition monitoring

Designation

Table 1: Designation example

Position																													
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	m	a	C	o	n	t	r	o	l	_	4	_	A	0	2	4	_	X	X	X	X	X	X	0	0	X	M	0	0

Table 2: Designation key

Position	Description	Code
1-10	Type series	
12	Design	
	L	Variant monitoring up to 3 sensors (⇨ Page 10)
	3	Variant monitoring up to 6 sensors (⇨ Page 10)
	4	Variant monitoring up to 6 sensors, current monitoring and Modbus RTU output (⇨ Page 10)
	5	Two-piece system design comprising an analysing unit and a sensor unit fitted directly in the submersible motor for monitoring up to 7 sensors, integrated vibration sensor, current monitoring and Modbus TCP output, incl. explosion protection approval (⇨ Page 10)
	X	Variant monitoring up to 3 sensors incl. explosion protection approval (⇨ Page 10)
14-17	Rated voltage [V]	
	A024	AC/DC 24 V Alternating current / direct current
	B230	AC 110 V - 240 V Alternating current
19	Analysing relay, motor temperature	
	B	Bimetal
	P	PTC thermistor
	X	Parameters not pre-set, can be set via the app
20	Analysing relay, temperature 1	
	B	Bimetal
	1	Pt100 resistance thermometer
	X	Parameters not pre-set, can be set via the app
21	Analysing relay, temperature 2	
	0	None

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Position	Description	Code
21	1	With Pt100 resistance thermometer
	X	Parameters not pre-set, can be set via the app
22	Analysing relay, resistance	
	0	None
	1	Float switch (mechanical seal leakage)
	X	Parameters not pre-set, can be set via the app
23	Analysing relay, conductive 1	
	1	Leakage sensor
	X	Parameters not pre-set, can be set via the app
	0	None
24	Analysing relay 4-20 mA	
	0	Without vibration monitoring
	1	With vibration monitoring
	X	Parameters not pre-set, can be set via the app
25	Analysing relay, temperature 3	
	1	Pt100
	X	Parameters not pre-set, can be set via the app
	0	None
26	Analysing relay, conductive 2	
	1	Leakage sensor
	X	Parameters not pre-set, can be set via the app
	0	None
27	Analysing relay, current measurement	
	1	Current measurement
	X	Parameters not pre-set, can be set via the app
	0	None
28	Field bus module	
	M	Modbus
	0	None
29	Communication input	
	M	Multi
	0	Single
30	Explosion protection	
	A	ATEX
	0	None

Functions

Depending on the design, the following functions are available:

- Monitoring of phase sequence, phase failure and asymmetry
- Analysis of overvoltages, undervoltages and short circuits
- Measurement of frequency of starts, operating hours and current

Various designs (⇨ Page 10)

Depending on the respective design, sensors can also be connected to monitor the following data:

- Temperature
- Conductance (leakage)
- Float switch
- 4-20 mA signal (vibration)

The warning values, setpoints and trip values can be individually parameterised via a digital interface and the app. The trip delay and re-start delay or interlocked tripping without automatic re-start can be set via the app. (⇨ Page 9)

Two parameterisable relays ensure the unit is reliably stopped or, respectively, a warning is emitted.

The diagnosis function provides operating data, fault lists, a fault counter, information on operating hours, start/stop cycles, current measured values, and a detailed fault analysis for the most recent fault (all measured values within a pre-defined time frame). At the digital interface the data can be read out on a smartphone, tablet or computer via the app. Depending on the design, information can be transmitted to an external system via Modbus RTU and/or Modbus TCP or by a gateway.

Configuration and function

The protection module is designed as an analysing unit and monitors external sensors. The plastic housing which is usually fitted in the control cabinet can be snap-mounted onto a 35 mm standard rail. Removable spring-loaded terminals provide the connection options for the power supply and the individual sensors. The respective actual values of the sensors are analysed and output.

Only AmaControl 5 comprises a sensor unit and an analysing unit. The external sensors are connected to the sensor unit. The signals from all sensors are transmitted digitally to the analysing unit via a 4-wire cable. AmaControl 5 cannot be purchased and/or retrofitted independently. Sensor units are factory-fitted in KSB submersible motors which are designed to accommodate them.

Diagnosis port (DP)

The diagnosis port (DP) is the data interface of AmaControl. This interface is used for setting parameters and/or reading out information. The KSB INTspector app is available for this purpose. AmaControl can be connected to a smartphone, tablet or computer via a USB gateway or Bluetooth gateway (accessories) (⇨ Page 9) .

The DP interface also provides the connection to the Modbus INT600 gateway and thus the output of data in the form of Modbus RTU.

LED flash code

The system condition can be analysed via the LED flash code . The flash code can also be entered in the INTspector app, which will display the corresponding message. The current message is directly displayed.

Modbus RTU interface

AmaControl 4 has an RS485 interface (screw terminal D0, D1, COM). This interface is used to forward information to higher-level systems and/or read it out.

Modbus TCP interface

AmaControl 5 has an RJ45 interface. This interface is used to forward information to higher-level systems and/or read it out.

Motor temperature monitoring

The motor temperature input is used to monitor the motor temperature. Depending on the design, this input can also be parameterised to PTC sensors, bimetal switches, Pt100 or Pt1000.

PTC

When the nominal trip temperature of the fitted sensors is reached, the sensors increase their resistance and trip the alarm relay without delay. A short circuit or open circuit at the temperature input will also trip the alarm relay.

Bimetal switch

When the nominal trip temperature of the fitted bimetal switch is reached, the bimetal switch opens and trips the alarm relay without delay.

Pt100 or Pt1000

If the input is parameterised to Pt100 or Pt1000, the temperatures for the warning and alarm limits must also be set.

Temperature monitoring

The temperature 1 and temperature 2 inputs monitor a temperature sensor, depending on the design. Depending on the design, this input can also be parameterised to bimetal, Pt100 or Pt1000.

Pt100

When the warning temperature and/or trip temperature of the fitted sensors (depending on the design) set in the protection device is reached, the alarm relay trips. A short circuit or open circuit at the temperature input will also trip the alarm relay. Re-starting takes place when the temperature falls below the trip temperature set in the protection device. The re-start conditions can be parameterised individually.

Leakage monitoring

The leakage 1 input (depending on the design) monitors a float switch in the leakage chamber of the mechanical seal.

Float switch: When the leakage chamber is full, the contact of the fitted float switch opens and trips the alarm relay after a pre-set trip delay. The trip delay can be set in the protection module. The protection module is re-activated without delay when the float switch contact closes again due to the leakage chamber being empty. The re-start conditions can be parameterised individually.

Conductance measurement

The leakage 2 input (depending on the design) monitors the electrodes in the winding space and connection space.

Electrodes (conductance measurement): When liquid enters, the ohmic resistance of the electrode measuring circuit drops. When the value falls below the warning resistance and/or trip resistance of the fitted electrodes set in the protection device, the warning relay and/or alarm relay are tripped. The trip delay can be set in the protection device. A short circuit or open circuit at the leakage input will also trip the alarm relay. Re-starting of the protection device takes place automatically and without delay (default setting) when the value exceeds the set trip resistance. A re-start delay or interlock can be parameterised in the protection device.

Vibration monitoring

The analog input (depending on the design) monitors a vibration sensor. The effective vibration velocity (root-mean-square value) of the sensor is read in the protection device as a 4 - 20 mA signal. In the protection device the values are also displayed in mm/s.

Vibration: When the warning limits and/or trip limits set in the protection device are reached, the warning relay and/or alarm relay are tripped. A short circuit or open circuit at the analog input will also trip the alarm relay. Re-starting takes place when the value falls below the value set in the protection device.

Vibration sensor (integrated in the sensor unit, AmaControl 5 only)

The vibration sensor is designed as a 3-axes vibration sensor and fitted directly on the printed circuit board of the sensor unit. In the protection device the values are also displayed in mm/s.

Vibration: When the warning limits and/or trip limits set in the protection device are reached, relay 1 and/or relay 2 may trip depending on the parameterisation.

Re-starting takes place when the value falls below the value set in the protection device. The re-start conditions can be parameterised individually.

Phase monitoring

Phase monitoring (L1, L2, L3, FE) is active when the motor has been started up; it monitors phase sequence, phase failure, phase asymmetry, undervoltage and overvoltage. When the motor is stopped, phase monitoring is disabled for approx. 2 seconds to prevent undesired tripping based on brief reverse rotation of the machinery. Phase monitoring can be set to “disabled” in the protection device.

The phase sequence is monitored. If the phase sequence is incorrect, the alarm relay trips. Re-starting takes place after the fault has been rectified and the protection device has been reset. Operating mode Phase sequence can be set to “disabled” in the protection device.

Phase failure is monitored the entire time the motor is running. The alarm relay is tripped when the value exceeds 75 % (default setting) of a phase based on the mean value of all 3 phases. Re-starting takes place automatically without delay (default setting) when the value falls below 75 %. The trip value in % and a re-start delay can be parameterised in the protection device.

Phase asymmetry is monitored the entire time the motor is running for a deviation in percent of a phase based on the mean value of all 3 phases. The warning relay or alarm relay are tripped without delay (default setting) when the value in % set in the protection device is exceeded. Re-starting takes place automatically and without delay (default setting) when the value falls below the trip value set in the protection device minus the pre-set hysteresis in %. The trip value, warning value, hysteresis, trip delay and re-start delay can be parameterised in the protection device.

Undervoltage and overvoltage are monitored the entire time the motor is running. When the warning limit (limit 1) or trip limit (limit 2) set in the protection device are reached, the warning relay or alarm relay are tripped. Re-starting takes place automatically and without delay (default setting), when the value falls below or rises above the trip value (limit 2) set in the protection device plus or minus the pre-set hysteresis. The operating mode, limit 1, limit 2, hysteresis, trip delay limit 1 and limit 2 as well as the re-start delay can be parameterised in the protection device.

Phase monitoring is also suitable for equipment operated on a frequency inverter from a frequency of 20 Hz.

Service interval function

The protection module has a pre-set service interval. When the configurable service interval has lapsed, the LED illuminates continuously or flashes during machine operation. The operating mode and interval can be parameterised in the protection module. The fitted LED signals the current status of the protection module .

Current transformer input

Current monitoring is single-phase by design via an external current transformer and is required to determine the power. The current of phase L1 can be monitored for overcurrent or undercurrent.

The current measurement is compatible with frequency inverters and a frequency range of 20 to 100 Hz. A specially developed current transformer is required for the measurement (⇨ Page 12) . Different current transformers are available, depending on the motor size.

A warning value and trip value as well as a hysteresis can be parameterised in the protection relay. When the set limit values are reached, the alarm relay or warning relay trips following the adjustable trip delay. A re-start delay or interlocked tripping can be parameterised for the alarm relay.

To fine-tune the measuring range and increase measuring accuracy, the number of primary winding turns of the cable passing through the current transformer can be set in the protection relay by parameter. The transmission ratio is determined by the current transformer.

An adjustable starting override interval during which monitoring is delayed after motor start-up prevents tripping at the moment of start-up.

A sensor error (no current transformer connected despite activated current monitoring, broken wire of current transformer, minimum current not reached during motor operation) leads to interlocked tripping of the alarm relay.

Current measurement is only possible when phases are connected to the protection relay.

Output relay/reset function

Trouble-free operation is signalled by the LED being lit in green. The alarm relay and the warning relay pick up. If a fault or warning are recognised, the alarm relay or warning relay open. The two output relays are of the normally closed type.

The alarm relay does not switch back until the fault has been rectified and a RESET has been carried out.

Table 3: Overview of reset functions

Type series	Reset functions		
	Power reset	External reset	Reset key
AmaControl 3	✗	-	-
AmaControl 4	✗	✗	-
AmaControl L	✗	✗	✗
AmaControl X	✗	✗	✗
AmaControl 5	✗	✗	✗

Connection options

AmaControl offers several options for reading out data:

- Directly on site
- Via Modbus in external systems

Directly on site

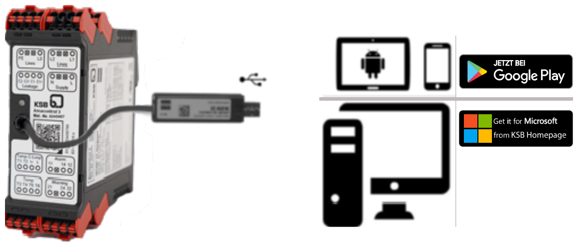

Fig. 1: Connection via USB

Fig. 2: Connection via Bluetooth

Further information:

- Accessories (⇨ Page 12)

External systems

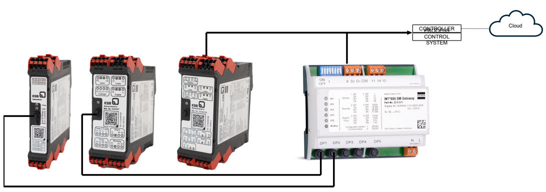

Fig. 3: Connection to a customer system or the cloud via Modbus gateway INT600 DM

Up to five AmaControl analysing units can be connected to the Modbus gateway INT600.

Further information:

- Accessories (⇨ Page 12)

Modbus RTU

AmaControl can make settings, operating data and recorded measured values available to a higher-level system (master) such as a control system via Modbus protocol (RTU). The protection relay acts as a slave by sending specific data values to the master after querying.

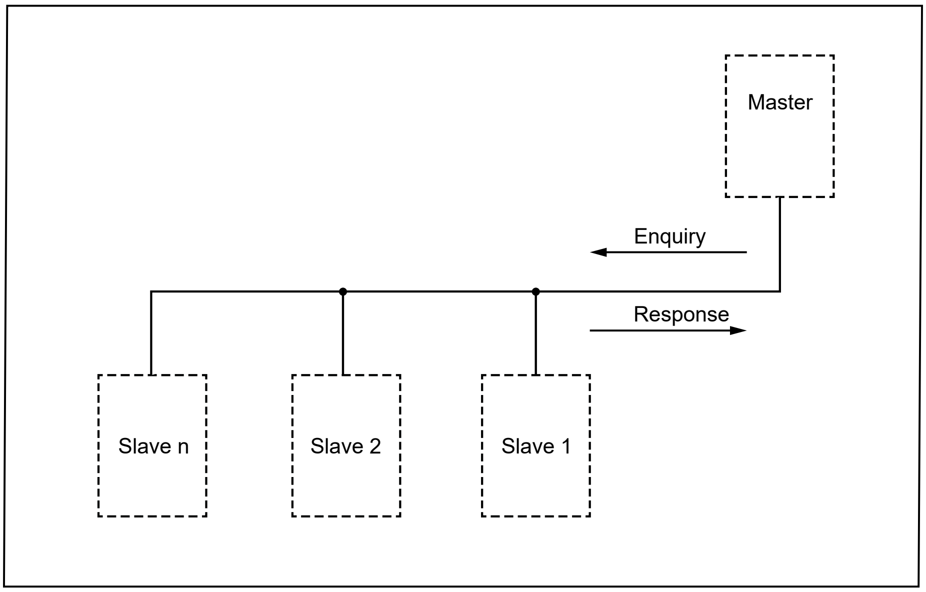

Fig. 4: Communication in a Modbus protocol network

Activity on the Modbus RTU can be viewed in the KSB INTspector app (separate LED not available).

Connections

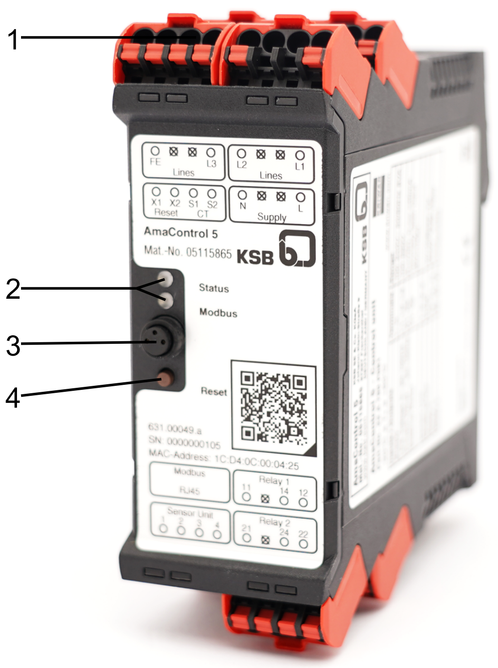

Fig. 5: Design of analysing unit for AmaControl 5 (example)

1	Terminals	2	LEDs
3	Diagnosis port (DP)	4	Reset key

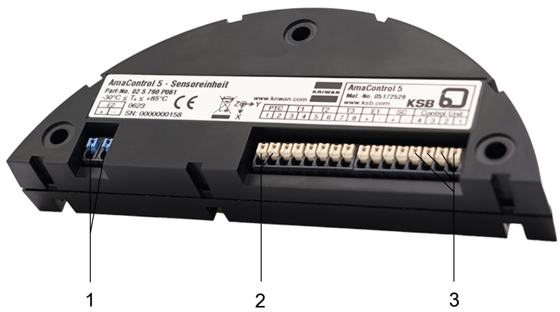

Fig. 6: Design of sensor unit for AmaControl 5 (example)

1	Sensor input with intrinsic safety barrier	3	Connection to analysing unit
2	Sensor inputs		

Connections

Depending on the respective design:

Inputs

- Sensor inputs
- Power supply N, L
- External reset
- Phase monitoring L1 - L3 (phase sequence, phase asymmetry, phase failure, undervoltage and overvoltage)
- Current transformer input S1, S2 for current measurement

The inputs have been set at the factory.

Outputs

- Relay 1 and relay 2 for warning and alarm
- LED(s)
- Data interface: diagnosis port (DP)
- Field bus module (Modbus)

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Available software/apps

AmaControl Select

Web app for selecting the right AmaControl for KSB products

https://www.ksb.com/static/Amacontrol-Select/

Available apps

KSB INTspector

This app allows parameters to be set and changed and stored data to be read out.

Download on the App Store

GET IT ON Google Play

Get it for Microsoft from KSB Homepage

Product benefits

- Protection and monitoring of pumps and submersible mixers by all-in-one device for motor temperature measurement, bearing temperature measurement, leakage measurement, vibration measurement and voltage measurement
- Easy to install in control cabinet on top hat rail; quick and easy wiring with spring terminals
- Straightforward commissioning as each variant is pre-parameterised for the product it will be used with
- Increased operating reliability due to warning and cut-out in the event of a fault, with warning option via general alarm contact
- Extended service life of pumps and submersible mixers by preventive maintenance information
- Simple analysis of diagnosis data with saved history, recorded operating data and trends as well as a fault history

Product information

Data access and use in accordance with Regulation (EU) 2023/2854 (EU Data Act)

AmaControl

Table 4: Device data recorded

Data type	Format	Volume	Continuous generation/ real-time generation	Data storage (articles 3, 2c)		Duration of retention
(Art. 3, 2a)	(Art. 3, 2a)	(Art. 3, 2a)	(Art. 3, 2b)	Device	Remote server	(Art. 3, 2c)
Motor temperature	Unsigned	16 bit	Yes	Yes	No	600 seconds
Bearing temperature, drive end	Unsigned	16 bit	Yes	Yes	No	600 seconds
Bearing temperature, pump end	Unsigned	16 bit	Yes	Yes	No	600 seconds
Leakage sensor	Unsigned	16 bit	Yes	Yes	No	600 seconds
Mechanical seal monitoring	Unsigned	16 bit	Yes	Yes	No	600 seconds
Vibration sensor	Unsigned	16 bit	Yes	Yes	No	600 seconds
Phase monitoring	Unsigned	16 bit	Yes	Yes	No	600 seconds
Frequency	Unsigned	16 bit	Yes	Yes	No	600 seconds
Undervoltage monitoring	Unsigned	16 bit	Yes	Yes	No	Instantaneous value
Overvoltage monitoring	Unsigned	16 bit	Yes	Yes	No	Instantaneous value
Frequency of starts	Unsigned	16 bit	Yes	Yes	No	Instantaneous value
Operating hours	Unsigned	32 bit	Yes	Yes	No	Instantaneous value
Event memory	Unsigned	16 bit	Yes	Yes	No	24 hours

Data access (articles 3, 2d)

Authorised users can access the collected data in the following ways:

- Data stored on the device
  - **Interface:** Bluetooth Low Energy (BLE), USB connection, Modbus RTU, TCP IP
  - **Technical details:** The customer is responsible for access rights, data storage and data deletion.
  - **Data format:** Exported data (e.g. vibrations, temperature, etc.) can be processed in commonly used tools (e.g. Excel). Internal data format (export as PDF): The KSB INTspector app is required for export.
- When an optional field bus interface is used, data may be stored on a **remote server** provided by the customer.
  - **Access:** The customer is responsible for managing, securing and updating the access data.
  - **Technical details:** The customer is responsible for access rights, data storage and data deletion.
  - **Data format**: Exported data can be processed in commonly used tools (e.g. Excel, ERP systems).

The requirements regarding the information obligation under the EU Data Act for related services such as the KSB ServiceTool and the KSB FlowManager are set out in a separate document. Only the data of the connected product itself is shown here.

Certifications

Table 5: Overview

Label	Effective in:	Comment
c UL us	USA, Canada	-
Ex	EU	Associated equipment for monitoring explosion-proof motors (AmaControl 5 and AmaControl X only).
FM APPROVED	USA	Associated equipment for monitoring explosion-proof motors (AmaControl 5 only).

Overview of product features / selection tables

Relevant software

AmaControl Select supports the selection of a suitable model for the product (⇨ Page 9)

Overview of product features

Table 6: Symbols key

Symbol	Description
D	Default setting
✗	Configurable via app
-	Not possible

Table 7: Overview of sensor inputs

Characteristic	AmaControl				
	3	4	L	X	5
Analysing relay, motor temperature					
Bimetal	✗	✗	D	-	-
PTC thermistor	D	D	✗	-	D
Pt100	✗	✗	✗	-	-
Pt1000	✗	✗	✗	-	-
Disabled	✗	✗	✗	-	-
Analysing relay, temperature 1					
Bimetal	-	-	✗	-	-
PTC thermistor	✗	✗	✗	D	-
Pt100	D	D	✗	-	D
Pt1000	✗	✗	✗	-	-
Disabled	✗	✗	D	-	✗
Analysing relay, temperature 2					
Bimetal	-	-	-	D	-
PTC thermistor	✗	✗	-	-	-
Pt100	✗	D	-	-	✗
Pt1000	✗	✗	-	-	-

Characteristic	AmaControl				
	3	4	L	X	5
Disabled	D	✗	-	-	D
Analysing relay, temperature 3					
PTC thermistor	-	-	-	-	-
Pt100	-	-	-	-	D
Pt1000	-	-	-	-	-
Disabled	-	-	-	-	✗
Analysing relay, leakage 1					
Float switch (mechanical seal leakage)	D	D	-	-	D
Disabled	✗	✗	-	-	✗
Analysing relay, leakage 2					
Leakage sensor or float switch	D	D	D	D	D
Disabled	✗	✗	✗	✗	✗
Analysing relay 4-20 mA					
With vibration monitoring	✗	✗	-	-	-
Disabled	D	D	-	-	-
Integrated vibration sensor					
3-axes vibration sensor	-	-	-	-	✗
Analysing relay, current measurement					
Current measurement	-	✗	-	-	✗
Field bus module					
Modbus TCP	-	-	-	-	✗[1]
Modbus RTU	✗[2]	✗[3]	✗[2]	✗[2]	✗[1]
ATEX certification	-	-	-	✗	✗

Selection table

Table 8: Overview of product features

Type series	Design		Mat. No.	Weight [kg]
	24 V	100-240 V		
AmaControl 3 A024 XXXXXX000000	✗	-	05040486	0.3
AmaControl 3 B230 XXXXXX000000	-	✗	05040487	0.3
AmaControl 4 A024 XXXX0X0XXM00	✗	-	05096165	0.3
AmaControl 4 B230 XXXX0X0XXM00	-	✗	05096166	0.3
AmaControl L A024 XX0X00000000	✗	-	05098601	0.3
AmaControl L B230 XX0X00000000	-	✗	05098602	0.3
AmaControl 5 A024 PXXXX1XXXTMA	✗	-	05115864	0.3
AmaControl 5 B230 PXXXX1XXXTMA	-	✗	05115865	0.3
AmaControl X A024 0PB0X000000A	✗	-	05098604	0.3
AmaControl X B230 0PB0X000000A	-	✗	05098603	0.3

Technical data

Table 9: Technical data

Characteristic	AmaControl 3/AmaControl 4	AmaControl L/AmaControl X	AmaControl 5
Rated voltage	AC/DC 50/60 Hz 24 V AC 50/60 Hz 100-240 V	AC/DC 50/60 Hz 24 V AC 50/60 Hz 100-240 V	AC/DC 50/60 Hz 24 V AC 50/60 Hz 100-240 V
Ambient temperature	-30 °C to +70 °C	-20 °C to +60 °C	-20 °C to +60 °C
Dimensions H×W×D	127 ×45×113 mm	127×23×113 mm	127×45×117 mm
Connections	Removable spring-loaded terminals	Removable spring-loaded terminals	Removable spring-loaded terminals

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1 Modbus TCP output via RJ45 interface or Modbus RTU output via INT600 DM Gateway (available as an accessory)
2 Modbus RTU output via INT600 DM Gateway (available as an accessory)
3 Modbus RTU output via RS485 interface or INT600 DM Gateway (available as an accessory)

Characteristic	AmaControl 3/AmaControl 4	AmaControl L/AmaControl X	AmaControl 5
Mounting	Snap-mounted on 35 mm standard rail	Snap-mounted on 35 mm standard rail	Snap-mounted on 35 mm standard rail
Interface	Diagnosis port	Diagnosis port	Diagnosis port
Indicator	LED for functional check via flash code	LED for functional check via flash code	LED for functional check via flash code

Accessories

Description	AmaControl					Mat. No.	[kg]
	3	4	L	X	5		
Connection cable (AmaControl - smartphone)	✗	✗	✗	✗	✗	01913080	0.1
Connection cable with Bluetooth interface (AmaControl - smartphone)	✗	✗	✗	✗	✗	01913079	0.1
Modbus INT600 DM gateway, supply voltage 230 V	✗[4]	✗[5]	✗[4]	✗[4]	✗[6]	01913082	0.2
Connection cable (AmaControl - gateway)	✗[4]	✗[5]	✗[4]	✗[4]	✗[6]	01913083	0.1
Current transformer 100 A for current measurement	-	✗	-	-	✗	05096163	0.15
Current transformer 200 A for current measurement	-	✗	-	-	✗	05096164	0.15

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4 Modbus RTU output via INT600 DM Gateway (available as an accessory)
5 Modbus RTU output via RS485 interface or INT600 DM Gateway (available as an accessory)
6 Modbus TCP output via RJ45 interface or Modbus RTU output via INT600 DM Gateway (available as an accessory)

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